# CHAPTER 7 ROTATION OF RIGID BODIES

**Key idea:** torque, moment of inertia and angular momentum play the roles of force, mass and momentum.

## 1 ROTATIONAL KINEMATICS

$$\omega = \omega_0 + \alpha t$$

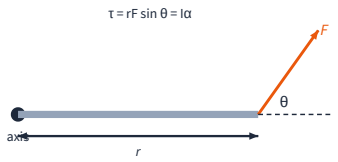
$$\theta = \omega_0 t + \tfrac{1}{2}\alpha t^2$$

$$\omega^2 = \omega_0^2 + 2\alpha\theta$$

Link to the rim:

$s = r\theta$ $v = r\omega$ $a = r\alpha$

## 2 TORQUE




$\tau = rF\sin\theta$ $\tau = I\alpha$

Unit N m; only the perpendicular component turns the body.

## 3 MOMENT OF INERTIA

$$I = \sum mr^2$$

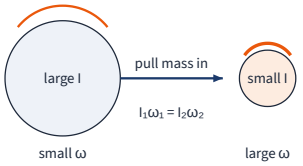
| Body (axis) | $I$ |
|---|---|
| Ring, hoop | $MR^2$ |
| Disc, cylinder | $\frac{1}{2}MR^2$ |
| Solid sphere | $\frac{2}{5}MR^2$ |
| Rod, centre | $\frac{1}{12}ML^2$ |
| Rod, end | $\frac{1}{3}ML^2$ |

Depends on mass, its distribution and the axis.

## 4 LINEAR ↔ ROTATIONAL

| Linear | Rotational |
|---|---|
| $s$ | $\theta$ |
| $v, a$ | $\omega, \alpha$ |
| $m$ | $I$ |
| $F = ma$ | $\tau = I\alpha$ |
| $p = mv$ | $L = I\omega$ |
| $\frac{1}{2}mv^2$ | $\frac{1}{2}I\omega^2$ |
| $W = Fs, P = Fv$ | $W = \tau\theta, P = \tau\omega$ |

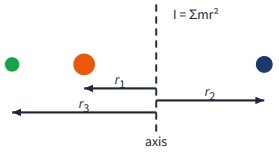
## 5 ANGULAR MOMENTUM




$L = I\omega$ $\tau = \dfrac{dL}{dt}$

**Conservation:** no external torque, $I_1\omega_1 = I_2\omega_2$ (skater pulls arms in: $I \downarrow, \omega \uparrow$).

## 6 ROTATIONAL KE & ROLLING




$$E_k = \tfrac{1}{2}mv^2 + \tfrac{1}{2}I\omega^2$$

Rolling without slipping: $v = r\omega$.

## 7 ROLLING DOWN A SLOPE FROM REST

$$mgh = \tfrac{1}{2}mv^2\left(1 + \frac{I}{mr^2}\right)$$

| Body | $v^2$ |
|---|---|
| Ring | $gh$ |
| Disc | $\frac{4}{3}gh$ |
| Sphere | $\frac{10}{7}gh$ |

Smaller $I/mr^2$ reaches the bottom first: sphere, then disc, then ring.

## 8 COMMON SECTION A TRAPS

- Using rev or degrees: convert to rad (1 rev = 2π rad).
- Forgetting the rotational KE of a rolling body.
- Using $I$ about the centre when the axis is at the end.
- Claiming kinetic energy is conserved when the skater pulls in (it increases).

## EXAM TIPS

- ✓Write $\tau = I\alpha$ for each rotating body and $F = ma$ for each hanging mass.
- ✓Link them with $a = r\alpha$.
- ✓Use angular momentum when the shape changes.
- ✓Use energy when heights are given.